

when is it your turn?

digitalisierung und programmierung · input, session 7

agenda

- 1 why it collapses
- 2 the clocked way
- 3 the clock-free way

why it collapses

how do eighty people
stay together?



your homework: where did it tip?

team

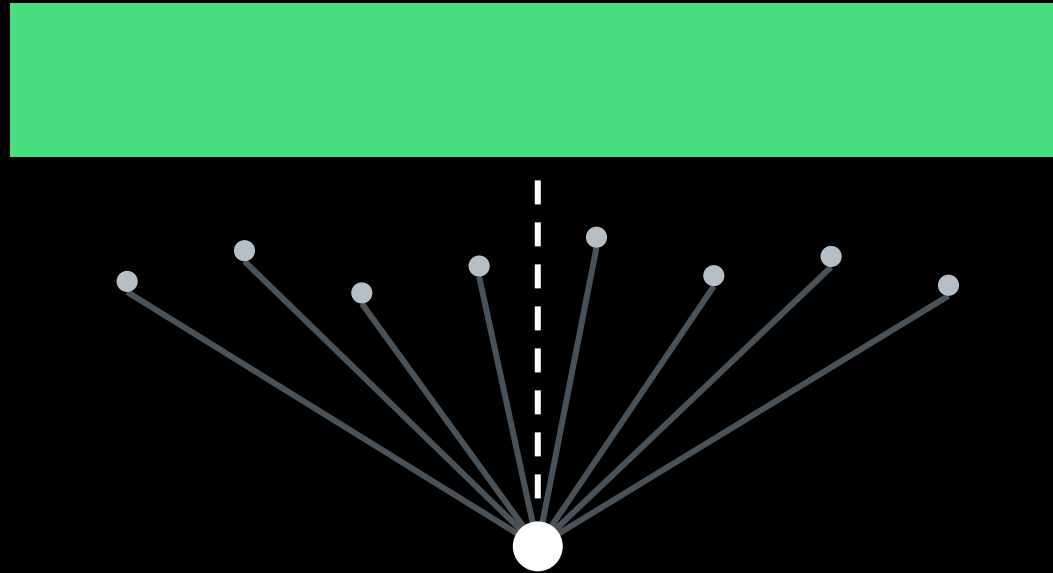
symbols per second

what broke

same limit for everyone. very different numbers.

what one dot means on the next slides

one symbol,
zoomed in

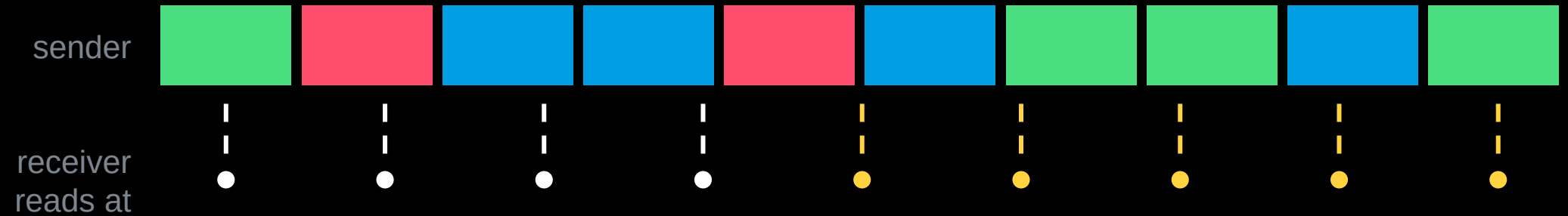


eight quick readings,
each a little different

their average: one value,
the dot on the next slides

one dot in the pictures = many readings, averaged.

cause one: no two clocks run alike



see it drift live:

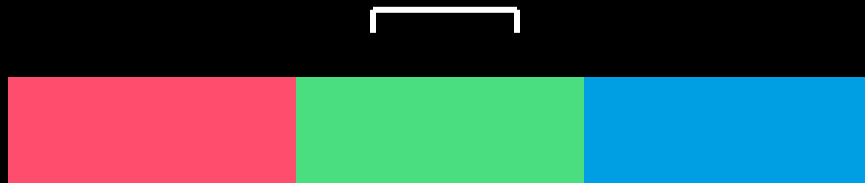
winf-hsos.github.io/lifi-concept-demos

drift adds up. the tiny error per symbol is not the problem, the sum is.

cause two: the sensor needs time to look

symbols 100 ms long

the sensor measures for 50 ms



it reports:



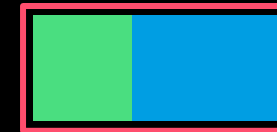
one colour, matches a profile

symbols 40 ms long

the sensor measures for 50 ms



it reports:



a blend of two, matches nothing

the sensor averages over its window. no error, no warning, just a meaningless number.

both causes are one question

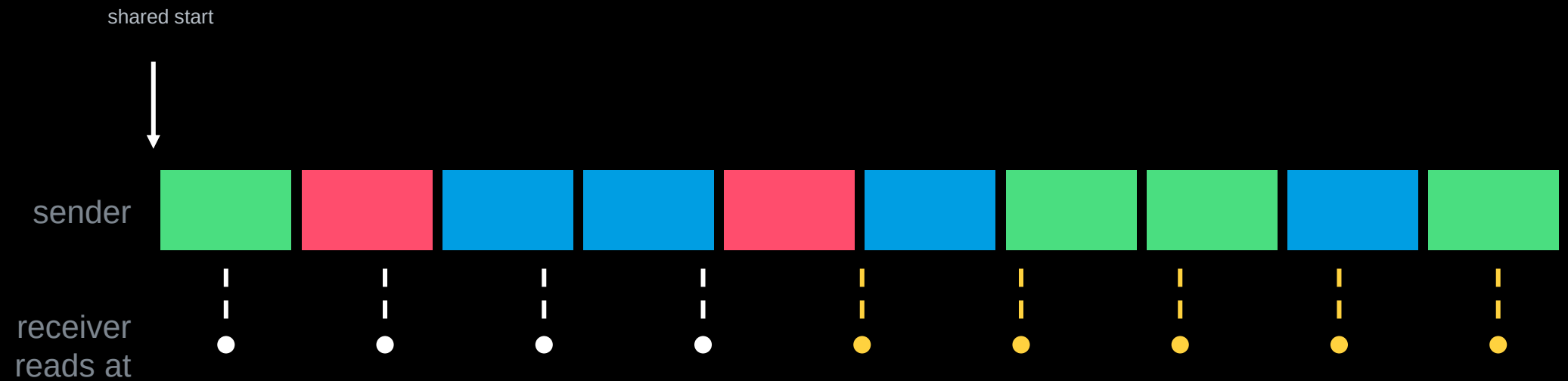
when does a new symbol begin?

answer one: time tells you.

answer two: change tells you.

the clocked way: time tells you

a shared start and a shared rate



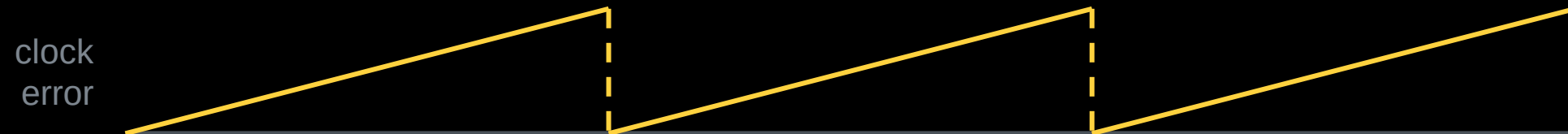
0.5 % clock difference at 10 symbols/s: off by one symbol after 20 s

a shared start is not a question of if it tips, only when.

recurring markers reset the drift



a marker is a symbol reserved for this job: a colour that never appears in your data (here: white), or a short dark pause. when the receiver sees it, it resets its clock.



the error can never grow beyond one block.

the price of the clocked way

start once

cheap: not a single symbol spent

drift adds up without limit

recurring markers

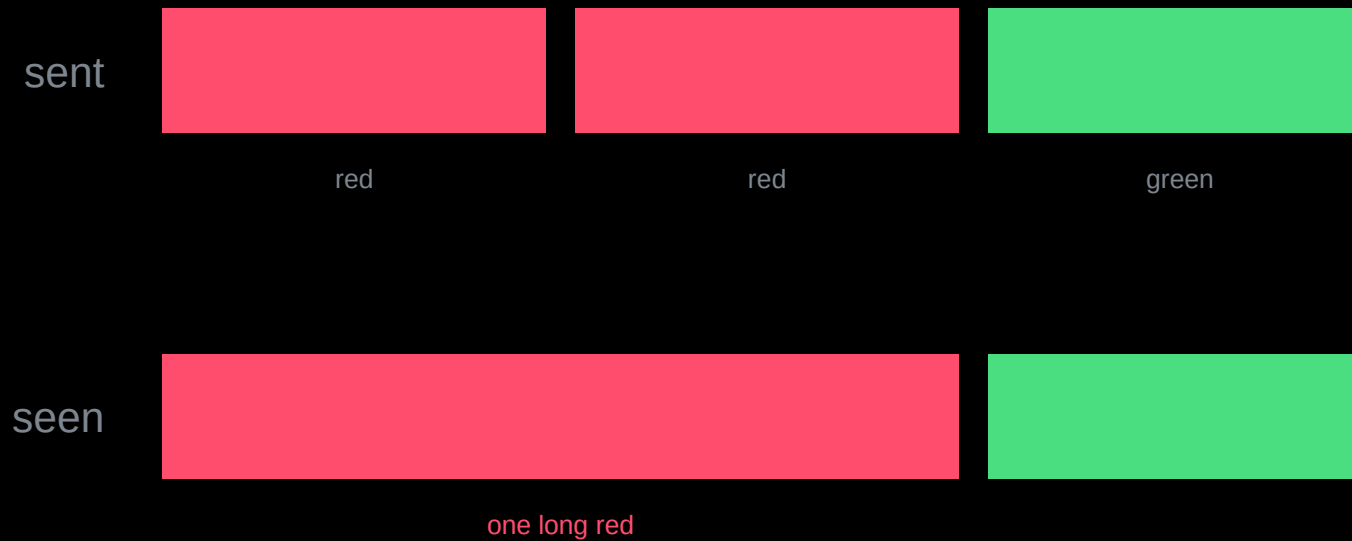
drift stays bounded

markers cost symbols that carry no message

how bad are your clocks? that is a measurement, not a belief.

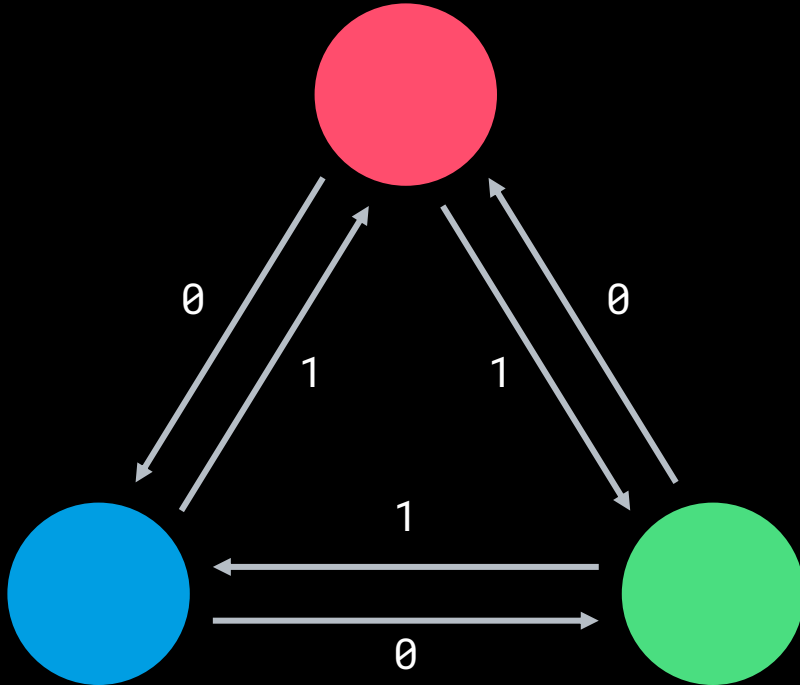
the clock-free way: change tells you

first idea: a change means „next symbol“

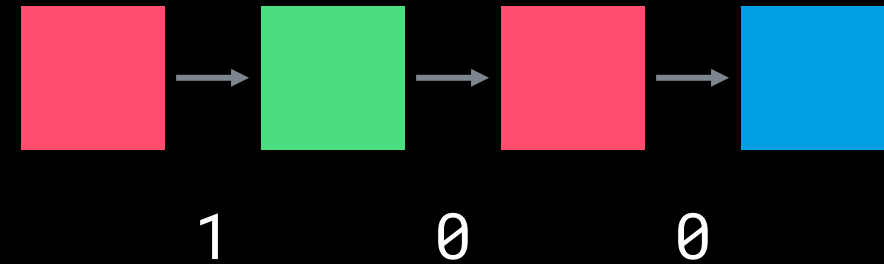


no change, no symbol: „red red“ arrives as one.

the fix: the transition carries the bit



example



measure as often as you like.
decode only on change.

your own colour is never allowed next.

freedom is paid in capacity

k colours as states

$\log_2(k)$ bits

k colours as transitions

$\log_2(k-1)$ bits

k = 3

1.58

1.00

k = 4

2.00

1.58

k = 5

2.32

2.00

want a clean 2 bits per transition? you need five colours, not four.

the most common wrong conclusion

clock-free is not time-free

how long must a colour hold to count?

how many transitions per second can the sensor follow?

what your file costs in minutes

$$t = \frac{n_{\text{bits}}}{r \cdot b}$$

n bits to send · r symbols per second · b bits per symbol

clocked: $b = \log_2(k)$

clock-free: $b = \log_2(k - 1)$

clocked, 16 000 bits, 4 colours, 5 symbols/s: about 27 minutes

two levers, nothing else: bits per symbol, symbols per second.

today

- 1 clapping experiment
- 2 workshop: find your maximum rate, with evidence
- 3 write down which way you lean: clocked or clock-free, and why